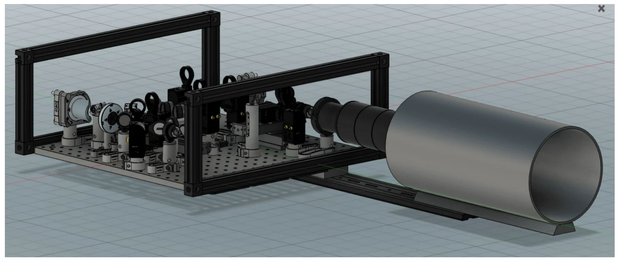
PANDORA: New avenues in sub-percent photometric precision for ground based astronomy

Kane M. Sjoberg, Johnny Esteves, Christopher W. Stubbs

Department of Physics, Harvard University

Introduction

Precision photometric calibration is key to a number of astrophysical research areas such as supernova cosmology and dark energy studies. In the age of large surveys, pushing the limits of photometry is a worthy challenge, with traditional celestial calibration methods limited to approximately 1% precision[1]. In this spirit we present PANDORA, a novel calibration source that achieves sub-percent photometric precision across the optical spectrum.



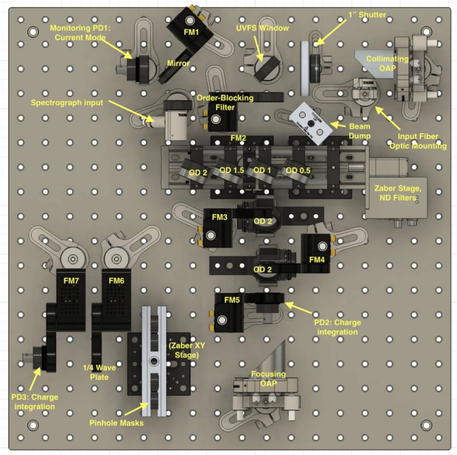
Engineering model of PANDORA.

PANDORA (Photometrically-Adjusted Neutral-Density Optical Ray Attenuator) is an experimental high-throughput design for in-dome photometric calibration of research telescopes. PANDORA delivers monochromatic (optical bandwidth <1 nm) doses of photons in the 400-1000 nm spectral range, enabling precise measurement of an instrument's throughput function.

The system is furthermore able to dynamically adjust the output beam intensity over six orders of magnitude, overcoming a limitation of designs employing integration spheres, which are inefficient and must overcome dynamic range mismatches with astronomical detectors.

Additionally, the system is self-monitoring during operation and is designed to perform routine internal throughput measurements for consistent self-calibration and reliability over time.

Methods

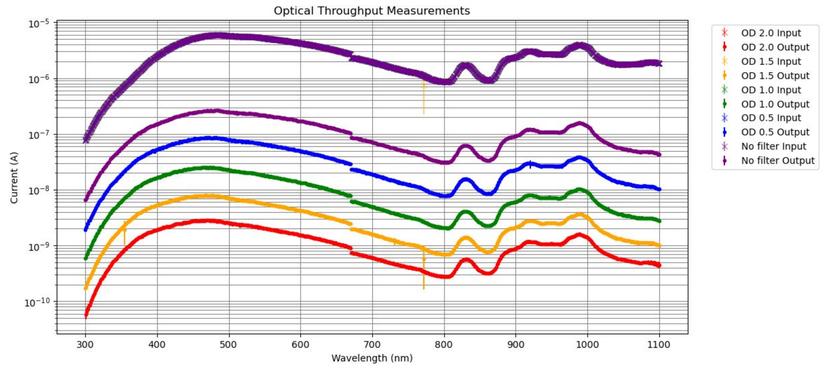


The 4th-generation optical design.

The system's light source is a temperature-stabilized, broadband Energetiq Laser-Driven Light Source (LDLS), which feeds into a Spectral Products CM112 Double Monochromator, delivering a wavelength-selectable output with sub-nanometer optical bandwidth. Attenuation is achieved through a selection of reflective neutral-density filters. The attenuated beam is then re-focused, masked in the focal plane, and projected through a 6" Ritchey-Chretien reflector. The whole optical payload is situated on an Alt-Azimuth telescope mount, enabling precise pointing control. The system's metrology standard is tied to NIST-calibrated photodiodes, which are read out on high-precision electrometers. All internal optical components are individually calibrated against these at sub-percent precision; the final output of the projector is additionally calibrated against a NIST-calibrated monocrystalline solar cell.

All system components are controlled from an onboard Mac Mini, interfaced via a Labjack and ethernet cables. It can be controlled remotely via command line and python scripts, enabling simple integration into existing observatory control systems for choreographed operation with the main telescope.

Results



Dark-calibrated wavelength vs. current plots from the high-flux monitoring photodiode "Input", and the downstream throughput-measuring photodiode "Output".

Shown above are current measurements acquired by the system with various ND filters inserted. The monitoring photodiode records high-current, high-SNR measurements, where the tight overlap indicates excellent light source stability across all wavelengths. Furthermore, the ND filters apply accurate attenuating factors to the photon flux passing through the system, indicating that performance is meeting desired throughput standards.

At the time of writing, PANDORA is undergoing final validation tests, and preparing to ship to Chile, where it will be calibrating the newly renovated ESO 1m Schmidt telescope, which will be performing a high-cadence sky survey in the *g,r,i,z* bands under the LS4 Collaboration (La Silla Schmidt Southern Survey).

Acknowledgements

The author would like to express their gratitude to Christopher W. Stubbs and Johnny Esteves for their time, mentorship, and support in the development of this instrument.

We are also grateful to the US DOE (DE-SC0007881 for DOE 133341), our LS4 collaborators, Schmidt Sciences, and Harvard University for their support.

1. Stubbs, C. W., Doherty, P., Cramer, C., et al. 2010, ApJS, 191, 376